



Analysis of Kenya's Agricultural Policy Environment Towards Attainment of Food Security and Climate Resilient Sector

Final Report

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ABBREVIATIONS

ASTGS	Agricultural Sector Transformation and Growth Strategy
CACA	County Assembly Committee on Agriculture
COMESA	Common Market for Eastern and Southern Africa
CSA	Climate Smart Agriculture
EAC-ARDP	East Africa Community Agriculture and Rural Development Policy
ECAS	Environmental Capacity and Sustainability Institute
FNSAP	Food and Nutrition Security Action Plan
UNGNA	United Nations Global Nutrition Agenda
IGAD	Intergovernmental Authority on Development
ILO	International Labour Organization
KJWA	Koronovia Joint Work on Agriculture
MOALF	Ministry of Agriculture, Livestock and Fisheries
NDC	Nationally Determined Contribution
NDMA	National Drought Management Authority
NEPAD	New Partnership for Africa's Development
PCA	Parliamentary Committee on Agriculture and Rural Development (PCA)
RECs	Regional Economic Communities
SDGs	Sustainable Development Goals
SMEs	Small Medium Enterprises
UNFCCC	United Nations Framework Convention on Climate Change (

EXECUTIVE SUMMARY

Policies have always played a big role in shaping sectoral operations and providing a road map to achieving particular targets within a sector. Kenya being an agro-economy, the sector has come up with several sectoral policies to guide in the achievement of particular targets related to the SDG2 and 13. The SDG 2 is geared toward zero hunger while SDG 13 is about climate action. This recognizes the fundamental importance of agriculture in responding to climate change and calls for joint work between the two Subsidiary Bodies of the United Nations Framework Convention on Climate Change. How far Kenya is in trying to achieve these two SDGs is the question that this paper is trying to answer. What policies are in place and how far the policies have pushed the agenda of zero hunger and climate action?

The study finds out that Kenya has several policies guiding the agricultural sector. These policies are at international level, regional level, country level as well as county levels. Kenya being a member state of the United Nations, there are a number of policies put in place by the UN to guide in achieving food security while guarding climate change. Some of these policies are the Kyoto protocol, Paris agreement COP 23 that brought to effect the Sustainable Development Goals.

At the national level, Kenya has developed a number of subsectoral policies that are geared toward providing a blue print to achieving zero hunger while promoting climate change action. Article 43 of the Constitution of Kenya provides for every person to have the right to be free from hunger and to have adequate food of acceptable quality. Article 53 of the constitution states that every child has the right to basic nutrition¹.

To consolidate laws on the regulation and promotion of agriculture, Kenya enacted the Agriculture, Fisheries and Food Authority Act, 2013 which establishes the Agriculture, Fisheries and Food Authority with respective roles of the national and county governments in agriculture excluding livestock and related matters and in furtherance of the relevant provisions of the Fourth Schedule of the Constitution.²

¹ See Constitution of Kenya, 2010

² The Agriculture, Fisheries and Food Authority Act, 2013

The Agriculture Act 2012 exists to promote and manage a stable agriculture, to provide for the conservation of the soil and its fertility and to stimulate the development of agricultural land in accordance with the acceptable practices of good land management and good husbandry³. Similarly, the Crops Act 2013 was established to provide for the growth and development of agricultural crops. To provide for the conservation, management and development of fisheries and other aquatic resources to enhance the livelihood of communities dependent on fishing, the Fisheries Management and Development Act, 2016 was enacted⁴. The Kenya Climate Smart Agriculture Strategy 2017-2026, is a tool to implement Kenya's Nationally Determined Contribution (NDC) for the agriculture sector and will require domestic and international support with a total of Ksh.500 billion (US\$ 5.0 billion) required for adaptation and mitigation actions for agriculture sector up to 2026. The Kenya Climate Smart Agriculture Implementation Framework 2018-2027 has been developed to promote climate resilient, low carbon growth and sustainable agriculture that ensures food security and contribute to national development goals in line with Kenya Vision 2030. The National Food and Nutrition Security Policy Implementation Framework 2017-2022 provides an overarching framework covering the multiple dimensions of food security and nutrition, and has been used to create synergy to existing sectoral and other initiatives of government and partners, including county governments.

To ensure the effective implementation of these policies, the government has come up with a number of institutions that are mandated to ensure the implementation of these policies. These range from the arms of the government to ministries, parastatals and committees that have been tasked to oversee different sectors and help achieve the government's targets.

³ The Agriculture Act 2012

⁴ The Fisheries Management and Development Act, 2016

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1. INTRODUCTION

Agricultural transformation is critical to growing the economy, reducing the cost of food, alleviating poverty and delivering 100% food and nutrition security. The ILO estimates that over 18 million Kenyans earn income from agriculture. The country's economic growth therefore depends on enabling these people to achieve food and nutrition security, and contribute more fully to the economy⁵. Article 43 (C) of the Kenya's Constitution grants all citizens the right to be free from hunger and have adequate food of acceptable quality.

According to the 2019 population census, total agricultural land stood at 10.3 million hectares which is equivalent to 17.5 per cent of the total country's land area. The census reports that 6.4 million were practicing agriculture, 5.6 million were growing crops, 369,679 were practicing irrigation, 5.1 million were engaged in maize cultivation, 3.6 million were cultivating beans, 4.7 million households were keeping livestock and 29,325 were in aquaculture as 109,640 households practiced fishing⁶. These farmers' output accounted for at least 63% of national produce, on approximately 90% of Kenya's land under agriculture.

Addressing climate change and sustainable natural resource management are at the heart of Kenya's ability to achieve 100% food and nutrition security, and there are risks involved when we fail to act. These risks include insufficient water basin management and poor irrigation practices characterized by uncontrolled abstraction of water and under exploitation of ground, storm, waste and saline waters; declining soil fertility because of poor nutrient management and farming practices reduces potential land output; limited support for climate-smart agriculture to increase agricultural productivity and incomes, adapt and build resilience to climate change, and reduce and/or remove greenhouse gas emissions; poor stewardship of fishing grounds which lowers the productivity of capture fisheries and aquaculture; and finally, insufficient disaster management due to inability to plan for emergencies. Addressing these challenges will not only increase

⁵ "Employment in agriculture (% of total employment) (modeled ILO estimate)", Employment to population ratio, 15+, total (%) (modeled ILO estimate)", and "Population, total", World Bank Development Indicators (2016), <http://databank.worldbank.org/>

⁶ Juliet Atellah – Kenyan Budget Allocation in the Sector of Agriculture for the FY 2020/2021

agricultural production sustainably and put food on the citizens tables, it will also ensure that future generations of Kenyans continue to benefit from agriculture.

Kenya has been experiencing increased drought after every two years and an increase in the intensity of each one whenever it strikes. Most of the land is arid and semi-arid with only around 17 per cent being of high or medium potential for intensive crop cultivation. The most shocking aspect is that the zones that were considered semi-arid are in danger of becoming arid whereas the arid regions have become so dry that no productive agricultural project can be successful. Only 7% of the cropped land in Kenya is under irrigation⁷.

According to KNBS⁸, whenever there is drought that lasts over six months, six million people sink back into poverty. This is an alarming issue which needs redress. These droughts are even affecting counties that were agriculturally viable and even classified as high yield zones which are now grappling with increased temperatures. Counties like Murang'a and Embu are examples of counties that produce tea and with the current trend it will be difficult to continue producing this high yield cash crops that help Kenya earn foreign exchange⁹.

Food security is one of the pillars within the big four Agenda and the National Climate Change Action Plan 2018-2022 (NCCAP II). This makes it prudent that to ensure that the current policies are updated or if need be new policies formulated to cushion the agriculture sector in Kenya. According to the Food and Nutrition Security Policy (FNSP), 2011, food and nutrition security is “a situation where all people, at all times, have physical and economic access to sufficient, safe and nutritious food to meet their dietary needs and food preferences for an active and healthy life”¹⁰. The policy states that the role of government is to ensure that all Kenyans enjoy safe food in sufficient quantity and quality to satisfy their nutritional needs for optimal health. Additionally, the UN Global Nutrition Agenda (UNGNA) states that nutrition security exists when food security is combined with education, a clean environment, adequate health services and proper care and feeding practices to ensure a healthy life for all household members.¹¹

⁷ P. Boulanger, H. Dudu, E. Ferrari, A.J. Mainar Causapé, J. Balié, L. Battaglia, Policy options to support the Agriculture Sector Growth and Transformation Strategy in Kenya. A CGE Analysis, EUR 29231, Publications Office of the European Union, Luxembourg, 2018, ISBN 978-92-79-85949-6, doi:10.2760/091326, JRC111251

⁸ See 2019 Kenya Population and Housing Census Reports

⁹ Ibid

¹⁰ The Food and Nutrition Security Policy (FNSP), 2011

¹¹ The UN Global Nutrition Agenda (UNGNA)

2. INTERNATIONAL AND REGIONAL POLICY LANDSCAPE

Kenya is a signatory to the 2030 Agenda for Sustainable Development whose Goal 2 emphasizes the need to end hunger, achieve food security and improved nutrition by the year 2030¹². In 2018, the agriculture negotiations under the United Nations Framework Convention on Climate Change (UNFCCC) successfully led to the adoption of decision 4/CP.23 on the Koronivia Joint Work on Agriculture (KJWA)¹³. The historic decision was adopted at the 2017 international climate conference, COP 23. The decision recognizes the fundamental importance of agriculture in responding to climate change and calls for joint work between the two Subsidiary Bodies of the United Nations Framework Convention on Climate Change. The six elements captured in the decision cover areas for action, including soil, livestock, nutrient and water management as well as the assessment of adaptation, socio-economic and food security dimensions¹⁴.

At the continental level, agriculture's share of employment and GDP is 61% and 25% respectively, yet 49% of the population live on less than a dollar per day. This implies that Africa's agriculture is not creating as much value as it should, compared to say Brazil and China that generate more wealth from fewer people. From the productivity dimension, the average yield for key staples and some export crops remain way below the global best practice.¹⁵ Africa's agriculture development is guided by the AUC-CAADP framework that commits member states to invest at least 10% of their annual budgetary allocations into agriculture¹⁶. It is implemented in the continent under the guidance of AUC-NEPAD and at the regional level by the Regional Economic Communities (RECs) who prioritize interventions towards food security. The RECS assist member states in developing viable national agriculture and food security investment plans, and implements policies and structures with CAADP framework. Kenya and other COMESA Member States signed a regional CAADP compact which identified four priority area for regional investments¹⁷:

¹² <https://www.undp.org/content/undp/en/home/sustainable-development-goals.html>

¹³ Koronivia Joint Work on Agriculture: Analysis of submissions, FAO

¹⁴ Ibid

¹⁵ Centre for Research on Financial Markets and Policy, 2018

¹⁶ AU 2003 Maputo Declaration on Agriculture and Food Security

¹⁷ <https://www.un.org/en/africa/osaa/peace/caadp.shtml>

1. Agricultural production and productivity with a focus on staple foods, livestock produce, fisheries and forest produce.
2. Removing barriers to agricultural trade and linking farmers to markets with a focus on corridors (corridor development).
3. Reducing social and economic vulnerability, and enhancing resilience and food and nutrition security.
4. Crosscutting issues including gender and age mainstreaming; human and institutional capacity development and strengthening; information and knowledge management; climate change; and improved coordination.

In East Africa, the EAC Agriculture and Rural Development Policy (EAC-ARDP 2006 and the EAC Food and Nutrition Security Action Plan (FNSAP) 2019-2023 serve as guiding documents. The EAC-FNSAP seeks to contribute to elimination of hunger, malnutrition, and extreme poverty in the East African region by the year 2023. The goal will be achieved through three interrelated objectives and six intermediate result areas as follows¹⁸:

- i. To improve sustainable and inclusive agricultural production, productivity and trade of crops, animal and animal resources, fisheries, aquaculture, apiculture and forest products;
- ii. To strengthen resilience among households, communities and livelihood systems by promoting sustainable utilization of natural resources, environmental conservation and uptake of disaster risk reduction with enhanced post-harvest value addition.
- iii. To improve access and utilization of nutritious, diverse and safe food.
- iv. The action plan has also proposed Cross-Cutting Intermediate Results (CCIR)

¹⁸ The East African Community Food and Nutrition Security Action Plan 2019-2023

3. NATIONAL POLICY AND LEGISLATIVE FRAMEWORKS

At the national level, Article 43 of the Constitution of Kenya provides for every person to have the right to be free from hunger and to have adequate food of acceptable quality. Article 53 of the constitution states that every child has the right to basic nutrition¹⁹. These articles were in response to the persistent food insecurity and drought episodes which made food security a major priority within government policy frameworks. Before 2010, the government had developed Vision 2030, a blueprint for the country's development agenda that captures the preferred economic growth trajectory. The Vision forms the basis of many policies, strategies and legislations²⁰. Vision 2030 identifies five strategic thrust areas for agricultural growth. These include:

- Transforming key institutions in agriculture and livestock to promote agricultural growth
- Increasing productivity of crops and livestock
- Introducing land use policies for better utilization of high and medium potential lands
- Developing more irrigable areas in arid and semi-arid lands for both crops and livestock
- Improving market access for the smallholders through better supply chain management.

The Vision aims at adding value to Kenya's farm and livestock products before they reach local and international markets. Food security is a prominent component of the Medium-Term Plans that operationalize Vision 2030.

In response to the need to transition Kenya's agriculture sector and make it a regional powerhouse, the government formulated the Agricultural Sector Transformation and Growth Strategy (ASTGS) based on the belief that food security requires a vibrant, commercial and modern agricultural sector that supports Kenya's economic development sustainably and its commitments to regional and global growth²¹. It seeks to contribute to national zero hunger strategic review and align with the government's Vision 2030, Third Medium Term Plan, the United Nations' Development Assistance Framework (UNDAF), the Sustainable Development Goals 2 and 17 as well as to

¹⁹ See Constitution of Kenya, 2010

²⁰ Kenya Vision 2030

²¹ The Agricultural Sector Transformation and Growth Strategy (2019-2029),

World Food Programmes Strategic Results 1, 4, 5 and 8.²² The ASTGS has three anchors to drive Kenya's 10-year transformation. The five-year target outcomes are highlighted in the table below:

Anchor	Target
Anchor 1	• Increase small-scale farmer, pastoralist and fisherfolk incomes • Raise average annual small-scale farmer incomes from KES 465/day to 625/day (~35% increase) • Directly benefit 3.3 million Kenyan farming households.
Anchor 2	• Increase agricultural output and value add • Expand agricultural GDP from KES2.9 trillion to KES3.9 trillion (6% CAGR). • Increase the contribution of agro- processing to GDP by KES130 billion over five years (50% increase over KES261 billion in 2018).
Anchor 3	• Boost household food resilience • Reduce the number of food-insecure Kenyans in the arid and semi-arid lands (ASAL) regions from an average 2.7 million to zero, while reducing the cost of food and improving nutrition. • Protect households against environmental and fiscal shocks.

Some of the flagship projects proposed in the ASTGS are shown in the table below:

Flagship	Target
Flagship 1	Target 1 million farmers in ~40 zones (initially) served by 1,000 farmer- facing SMEs

²² <https://theexchange.africa/industry-and-trade/kenya-agriculture-budget-allocation-estimates-maputo-declaration-food-security/>

Flagship 2	Shift nationwide subsidies focus to register 1.4 million high-needs farming households and empower them to access a range of inputs from multiple providers through e-vouchers
Flagship 3	Establish 6 large-scale agro- processing hubs through a one-stop shop for agro-processors
Flagship 4	Unlock 50 new large-scale private farms (bigger than 2,500 acres) and sustainable water supply for more than 150,000 acres of irrigation from existing infrastructure
Flagship 5	Restructure governance and operations of the Strategic Food Reserve (SFR) to better serve 4 million vulnerable Kenyans
Flagship 6	Boost the food resilience of 1.2 million farming and pastoralist households in arid and semi-arid lands (ASALs) through community-driven intervention design
Flagship 7	Launch three skill programmes for 200 government leaders, flagship implementers and 3,000 youth-led and digitally-enabled extension agents
Flagship 8	Strengthen research and innovation as launch priority digital and data use cases to drive better decision-making and performance management
Flagship 9	Monitor two key food system risks – those addressing sustainability and climate, and a second category for crisis management for pests, diseases and global price shocks.

To consolidate laws on the regulation and promotion of agriculture, Kenya enacted the Agriculture, Fisheries and Food Authority Act, 2013 which establishes the Agriculture, Fisheries and Food Authority with respective roles of the national and county governments in agriculture excluding

livestock and related matters and in furtherance of the relevant provisions of the Fourth Schedule of the Constitution.²³

The Agriculture Act 2012 exists to promote and manage a stable agriculture, to provide for the conservation of the soil and its fertility and to stimulate the development of agricultural land in accordance with the acceptable practices of good land management and good husbandry²⁴. Similarly, the Crops Act 2013 was established to provide for the growth and development of agricultural crops. It is purposed to accelerate the growth and development of agriculture, enhance productivity and incomes of farmers and the rural population, improve investment climate and efficiency of agribusiness and develop agricultural crops as export crops that will augment the foreign exchange earnings of the country, through promotion of the production, processing, marketing and distribution of crops in suitable areas of the country. Particularly, it is expected to circumvent unnecessary levies, taxes or other barriers to free movement of crops products and provide for a rationalized taxation system; reduce unnecessary regulation or overregulation of the crops subsector; reduce duplication and overlap of functions among institutions involved in the regulation of crop agriculture; promote competitiveness in the crops subsector and to develop diversified crop products and market outlets; and attract and promote private investment in crop agriculture²⁵.

To provide for the conservation, management and development of fisheries and other aquatic resources to enhance the livelihood of communities dependent on fishing, the Fisheries Management and Development Act, 2016 was enacted²⁶. The objective of the Act is to protect, manage, use and develop the aquatic resources in a manner which is consistent with ecologically sustainable development, to uplift the living standards of the fishing communities and to introduce fishing to traditionally non-fishing communities and to enhance food security.

The Kenya Climate Smart Agriculture Strategy 2017-2026, is a tool to implement Kenya's Nationally Determined Contribution (NDC) for the agriculture sector and will require domestic and international support with a total of Ksh.500 billion (US\$ 5.0 billion) required for adaptation and mitigation actions for agriculture sector up to 2026. These actions are expected to contribute

²³ The Agriculture, Fisheries and Food Authority Act, 2013

²⁴ The Agriculture Act 2012

²⁵ Crops Act 2013

²⁶ The Fisheries Management and Development Act, 2016

to building resilience and adaptive capacity in the sector as well as reduce sectoral emissions to 30 MtCO₂e relative to the business as usual trajectory projection of 37 MtCO₂e in 2026. The strategy provides a detailed implementation framework with clear stakeholder roles and responsibilities and forms a basis for the establishment of a monitoring and evaluation (M&E) framework. The Strategy defines climate smart agriculture (CSA) as an approach that helps to guide actions needed to transform and reorient agricultural systems to effectively support development and ensure food security in a changing climate. CSA aims to achieve three main objectives: sustainably increasing agricultural productivity and incomes; adapting and building resilience to climate change; and reducing and/or removing greenhouse gas emissions, where possible. These objectives form part of Kenya's obligation as a signatory to the United Nations Framework Convention on Climate Change (UNFCCC)²⁷.

The Kenya Climate Smart Agriculture Implementation Framework 2018-2027 has been developed to promote climate resilient, low carbon growth and sustainable agriculture that ensures food security and contribute to national development goals in line with Kenya Vision 2030. The Framework has been developed to set guidelines for implementing Climate Smart Agriculture (CSA) approaches, strategies, practices and technologies in Kenya. The guidance will be at both National and county levels and above all in the context of global obligations and local realities in consideration that Kenya is a signatory to the Kyoto protocol and the Paris Agreement. The document is realigned to the need for Kenya to realize its development goals and leverage on global opportunities by providing clear structures through which farmers can benefit from tapping the available global climate financing instruments in collaboration with other players²⁸.

The National Seed Policy 2010 outlines the intervention measures the government pursues in research, extension, seed production, multiplication, processing, marketing and distribution. The Policy is intended to avail adequate high-quality seed and planting material for the country's farming and forestry needs.²⁹

The National Food and Nutrition Security Policy (FNSP) 2011 provides an overarching framework covering the multiple dimensions of food security and nutrition improvement. It has been

²⁷ The Kenya Climate Smart Agriculture Strategy 2017-2026

²⁸ The Kenya Climate Smart Agriculture Implementation Framework 2018-2027

²⁹ The National Seed Policy 2010

developed to add value and create synergy to existing sectoral and other initiatives of government and partners. It recognizes the need for multi-public and private sector involvement, and that hunger eradication and nutrition improvement is a shared responsibility of all Kenyans. The policy is a framework in the context of basic human rights, child rights and women's rights, including the Universal Right to Food. It also entrenches the government's policy that all Kenyans, throughout their life-cycle enjoy at all times, safe food in sufficient quantity and quality to satisfy their nutritional needs for optimal health. It addresses associated issues of chronic, poverty-based food insecurity and malnutrition, as well as the perpetuity of acute food insecurity and malnutrition associated with frequent and recurring emergencies, and the critical linkages with specific focus towards: food availability and access; food safety, standards and quality control; nutrition improvement; school nutrition and nutrition awareness; food security and nutrition information; early warning and emergency management; institutional and legal framework and financing; strategic approaches for policy implementation, monitoring and evaluation³⁰.

The National Food and Nutrition Security Policy Implementation Framework 2017-2022 provides an overarching framework covering the multiple dimensions of food security and nutrition, and has been used to create synergy to existing sectoral and other initiatives of government and partners, including county governments. The Policy covers all the four dimensions of food security (including availability, accessibility, stability and meeting nutritional requirements) and combines long-term action to enhance productivity and incomes, and respond to immediate needs of the poor and food insecure. The broad objectives of the FNSP are:

- To achieve good nutrition for all Kenyans
- To increase the quantity and quality of food available, and ensure food is accessible and affordable to all Kenyans.
- To protect vulnerable populations using innovative and cost-effective safety nets linked to long-term development.

Kenya Youth in Agribusiness Strategy 2017-2021 has been developed to address challenges that hinder youth from participating effectively in the sector. The Strategy is aimed at providing new opportunities for youth in agriculture and its value chains. The impact of youth engagement in

³⁰ The National Food and Nutrition Security Policy (FNSP) 2011

agriculture is expected to manifest in sustainable economic growth and in the reduction of poverty and malnutrition. This will ensure optimal utilization of youth potential in contributing to the sector goals of achieving food and nutrition security, income generation, employment and wealth creation in the country. The Strategy has identified strategic issues which include negative perception to agricultural activities, large population of youth have inadequate skills, knowledge and information; limited participation of youth in agricultural innovations, research, technology development and utilization; limited access to land for agribusiness, limited access to financial services, limited access to market information; inadequate market infrastructure and entrepreneurship; negative impacts of climate change and weak environmental governance; low levels of value addition; limited access to market information; inadequate³¹.

The National Agribusiness Strategy 2012 also strives to make Kenya's agribusiness sector a competitive driver of growth. It identifies, analyses and overcomes the systemic bottlenecks that stand in the way of realizing Kenya's enormous entrepreneurial potential³². It proposes how to introduce the systems and structures needed to bring about a dynamic and competitive agribusiness sector in Kenya. It aims to make existing systems work more flexible and adaptively to suit changing conditions and in a way that can exploit new market opportunities. The Strategy proposes how the agribusiness sector can play a key role in realizing Kenya's 2030 vision of annual economic growth rates of 10% from 2012. The strategy, if successful, will deliver wealth creation, job creation and food security within national borders, even in a context of rapidly changing climate and environmental degradation. Importantly, it is expected to feed into other economic sectors in vicious circle as well as contribute significantly and positively towards the fulfillment of all the six pillars of the Medium-term Investment Plan (MTIP) of the Agriculture Sector Development Strategy (ASDS).

The National Agricultural Research System Policy 2012 seeks to streamline, rationalize and put in a system that is consultative, efficient and effective and takes into account economics of scale to not only use the current scientific, human and physical capacities but also position Kenya as a hub for agricultural research and development in the region. It is aimed at creating an enabling

³¹ Kenya Youth in Agribusiness Strategy 2017-2021

³² National Agribusiness Strategy 2012

environment for a agricultural research industry that contributes to the overarching national policies of economic growth and wealth creation, poverty reduction and gender equity.

The National Agricultural Sector Extension Policy (NASEP) 2012 spells out modalities for effective management and organization of agricultural extension in a pluralistic system where both public and private service providers are active participants. The Policy provides a point of reference for service providers and other stakeholders on standards, ethics and approaches, and guides all players on how to strengthen coordination, partnership and collaboration. It adopts a sector-wide approach to providing extension services. It is to enable the extension service to realize its vision which is: Kenyan agricultural extension clientele demand and access appropriate quality extension services from the best providers and attain higher productivity, increased incomes and improved standard of living³³.

The Ministry has attempted to build up a comprehensive, reasonable and practical National Rice Development Strategy (NRDS Stage 2) for the period 2019-2030. The strategy centers around what actors plan to achieve and how assets will be coordinated towards understanding the ideal objectives inside the given time span. The NRDS focuses to address numerous challenges faced by stakeholders in the value chain and offers various opportunities for increased productivity and production³⁴. The challenges include: costs of production, erratic weather and water supply, lack of credit facilities, high harvest and post-harvest losses, poor infrastructure in irrigated ecologies, and poor market and market infrastructure among others. The NRDS stage 2 will expand on accomplishments made in stage 1 which included utilization of hybrid breed seeds, enhancement of the Water Saving Rice Culture, enhanced mechanization along the rice value chain, advancement, what's more, the release of improved assortments, the development of rice seed dispersion framework, capacity building for staff and ranchers, development of two rice research labs and improved networking among others.

When Jubilee government retained their contract to rule Kenya for the next 5 years, they designed the Big Four Agenda³⁵, that incorporates 100% food security and nutrition through new and innovative initiatives over the five years as detailed in the table below:

³³ National Agricultural Sector Extension Policy

³⁴ https://riceforafrica.net/images/countries/NRDS_rev/NRDS2_Kenya_en.pdf

³⁵ <http://cn.invest.go.ke/wp-content/uploads/2018/12/Government-of-Kenya-Big-Four-Plan.pdf>

No	Focus Area	Detailed Initiatives
1.	Enhance large scale production	- Place additional 700,000 Acres through PPP (including idle arable land) under maize, potato, rice, cotton, aquaculture and feeds production. - Form an Agriculture and Irrigation Sector Working Group (AISWAG) to provide coordination for irrigated Agriculture. - Use locally blended fertilizer on a 50/50 basis and implement liming. - Avail incentives for post-harvest technologies to reduce postharvest losses from 20% to 15% e.g. waive duty on cereal drying equipment, hematic bags, grain cocoons/silos, fishing and aquaculture equipment and feed.
2.	Drive small holder productivity	- Establish 1,000 targeted production level SMEs using a performance-based incentive model in the entire value chain - Improve access to credit/input for farmers through Warehouse Receipt System and strengthen commodity fund - Establish commercialized feed systems for livestock, fish, poultry and piggery to revolutionize feed regime and traceability of animals - Establish East Africa's Premier food hub, secure investors to construct a Shipyard (in 2018 – site existing) and increase domestic fishing fleet by 68 Vessels in the Coast.
3.	Reduce cost of food	- Contract farmers for Strategic Food Reserve and other commercial off-takers - Redesign subsidy model to maximize impact by focusing on specific farmer needs (flexible voucher and incentive-based model) - Secure investments through PPP in post-harvest handling (storage, cold storage for fish, aggregation) and market distribution infrastructure to reduce losses (by Dec 2018) - Eliminate multiple levies across counties in the agriculture value chain (enforce laws on roads)

4. POLICY IMPLEMENTATION- INSTITUTIONAL ARRANGEMENTS

The devolved system of government has resulted in significant changes in government structures and organization. The constitution created one national government and 47 semi-autonomous county governments. This has resulted into most agricultural functions being transferred to county level, which has significantly impacted the policy formulation and implementation process.

Kenya's governance system consists of the three arms of government: the executive, parliament and judiciary. Policy development and related national legislations is a shared function of the executive branch and the legislature at the national and county governments. Each ministry is responsible for policy issues under its docket and consults with the attorney general's office and treasury on legal or financial matters. A policy document is presented to the relevant parliamentary committee and subsequently to the cabinet through a memo. It is ultimately presented to the National Assembly for discussion and passed as erosional paper. Both levels of government develop legislations and regulations where the process is like policy development at the national level. This is replicated at the county level through the county assemblies³⁶.

Regarding agricultural policies, the national government has developed many accompanying legislations and regulations. Moreover, county governments are developing county-based legislations and regulations that are relevant to agriculture such as specific crop and livestock regulations. Food and nutritional security is a multi-sector and multi-stakeholder concern that requires concerted efforts but the stakeholders have no forum for coordination to ensure seamlessness on the matter.

To start with, the Ministry of Agriculture, Livestock and Fisheries (MOALF) at the national level is responsible for agriculture, livestock and sustainable fisheries policies, while the county governments are responsible for implementation of the agriculture and veterinary policies. The table³⁷ below highlights the institutional arrangements supporting the agriculture sector:

No	Institution	Role/Mandate
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³⁶ KIPPRA, Public Policy Making Process in Kenya

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1.	Department of Livestock:	Responsible for livestock and veterinary policy.
2.	Department of Fisheries	Responsible for fisheries policy
3.	Department of Agriculture:	Responsible for crop policy and food security policy.
4.	Ministry of Water and Irrigation	Responsible for irrigation policies and strategies and their implementation
5.	State Department Cooperative	Is responsible for policies and regulations on all small and large farmer cooperative movements
6.	Ministry of health	Is responsible for matters of nutrition
7.	Kenya Agriculture and Livestock Research Organization	Is responsible agricultural research
8.	Intergovernmental Secretariat (IGS)	Created under the Intergovernmental Relations Act 2012 (IRA) in order to promote intergovernmental relations. It is meant to facilitate and improve coordination in the agricultural sector.
9.	The Summit	This is a consultative body that brings together the COG and the national government to deliberate on issues that arise during the planning period. The role of the summit is to support strategies that result from sector- related consultative forums, and strengthen coordination between the two levels of government. For the sector issues to feature in the Summit, the lower level sector structures at both levels will require capacity for internal processes and for relationships with other actors to influence the agenda of the Summit to support agriculture policy reform process.

10.	Council of Governors (COG)	The Council of Governors is the apex body of the 47 county governors. The council identifies priority issues and collectively addresses issues of public policy and governance at the county and national levels. The vision of the council is to have prosperous and democratic counties delivering services to every Kenyan. The mission of the council is to benchmark excellence in devolution that is non-partisan; providing a supporting pillar for county governments as a platform for consultation, information sharing, capacity building, performance management and dispute resolution ¹ . The council has 10 technical committees that are mainly sectoral-based to effectively achieve desired results.
11.	Agriculture Committee (AC) of the Council of Governors	The council addresses all matters related to agriculture and food security including the following: i) sustainable agricultural practices; ii) poverty eradication iii) value addition for farmers; iv) food security and drought management; v) production and marketing; vi) fisheries development; and vii) adoption of technological advancements in agriculture. The committee also addresses matters related to land survey and mapping, and seek out sustainable and accountable solutions for subdividing and transferring land.
12.	Parliamentary Committee on Agriculture and Rural Development (PCA)	The National Assembly has formed sector committees to enhance efficiency and effectiveness. The role of NACA is to advise the parliament on all issues related to agriculture. This committee is involved actively in the agriculture policy process.

13.	Senate Committee on Agriculture (SCA)	The Senate has formed a committee on agriculture to address all matters related to policy and regulations at the county level. Currently, the committee is involved in policy formulation processes as demonstrated by their participation in the recently drafted agriculture policy. The SCA and NACA have a structured working relationship and often hold discussions on policy-related issues in agriculture sector. Their discussions have enhanced citizens' engagement in the policy process. They also provide an oversight role in the implementation of other sector policies
14.	Caucus for CEC members for Agriculture	This group is composed of all 47 CEC members for agriculture and consists of an executive arm and four technical thematic groups. The CEC Agriculture Forum is part of the IGS at the national level and addresses county policy issues. Five members of the forum represent the counties at the IGS. The committee has not been able to fulfill their role effectively due to constraints related to internal processes at the national level, and the commitment of the two levels of government to engage in dialogue and consultations regarding the policy process.
15.	The County Assemblies Forum (CAF)	This forum is the coordinating body of the 47 county assemblies. The forum seeks to strengthen capacity and institutionalize law-making for the county assemblies. It also seeks to form linkages with other arms of government. Insert the role of CAF in food security

16.	County Assembly Committee on Agriculture (CACA)	The county assemblies (CAs) are responsible for drafting laws within their respective counties. In order to achieve its objective, CAs often have committees that deliberate on sectoral matters and guide the assemblies. The County Assembly Committee on Agriculture is a committee that assists CAs on all matters related to agriculture and food security policy. Their participation in the national policy reform process is very important for the integration of national policies into county government processes. Although this committee significantly participates in policy processes, the county government still exhibits capacity challenges.
17.	County Executive Committee for Agriculture	This committee coordinates agricultural policy implementation at each county. It is headed by the CEC for agriculture and composed of technical directors of multiple departments. Its major role is to implement sector policies at the county level. The composition of this committee varies from county to county depending on what constitutes the agricultural sector.
18.	Agriculture and Rural Development (ARD)	A donor working group formed to engage development partners in the overall policy process. The group supports the government by providing information on international best practices and supporting policy reforms. Moreover, they provide substantive finances and technical assistance to implement the policies through projects and programs. Challenges that hinder the effectiveness of the working groups include inadequate coordination within groups and among different actors

19.	Research Institutions	Agriculture research is guided by the National Agriculture Research Systems Policy (NARSP). The main objective of NARSP is to provide direction for national research and sustainable development. This involves the formulation of research agendas to focus on impact-driven national priorities consistent with sectoral policies and strategies. In addition, Kenya is home to several international, regional research institutions among them CYMMIT, ICRISAT, ICRAF. In addition to the local research institutions there are international institutions that collaborate with National research institutes
20.	Universities	Universities build the capacity of the agriculture work force and participate in the collection of data to inform policy dialogue. They also do research in support of food and nutrition security. In the formulation process, they provide staff for analysis both at field level and at the policy consultancy and advisory level
21.	Joint Agriculture Sector Consultation and Cooperation Mechanism (JASCCM)	was established under the Intergovernmental Relations Act 2012 and aims to enhance consultation and cooperation between the two levels of government. It has three layers of functional responsibilities: • The Intergovernmental Forum for Agriculture (IGF); • Joint Agriculture Steering Committee (JASSCOM); and • Sectoral working groups. The mechanism is served by the Joint Agriculture Secretariat (JAS). The mechanism facilitates sector consultations and communication. JAS has consequently

		developed a joint sector work plan and a code of conduct for partners during the implementation of sector programs and projects.
22.	The Agriculture Council of Kenya (AgCK)	This is the apex body for all agriculture sector private stakeholders. The private sector includes all value chain actors whose interest are commercial gains and includes producers, input suppliers, agro-traders (retailers and wholesalers) and agro-processors. Due to the diversity of agricultural commodities, organizing private sector actors to participate effectively in policy formulation has remained a significant challenge to coordinators. However, with regard to implementation, they are effectively engaged. Recently, the private sector has established a mechanism to actively engage in the policy processes.
23.	Civil Society Organizations (CSO)	have focused more on service delivery within the sector. A limited amount of CSOs in the sector participate in policy formulation, including KENAFF and LMO. A major challenge to their effective participation in the policy process is a lack of capacity in policy advocacy.
24.	Community Based Organizations (CBO) and Faith Based Organizations (FBOs)	CBOs are nonprofit groups that work at a local level to improve life for residents. They cover a series of aimed at bringing about desired improvement in the social wellbeing of individuals, groups and neighborhoods. A Faith-based organization is an organization whose values are based on faith and/or beliefs, which has a mission based on social values of the particular faith, and which most often draws its activists from a particular faith group.

5. DEVOLVING AGRICULTURE AND FOOD SECURITY IN KENYA

Pursuant to the Fourth Schedule of the Constitution, the county governments are responsible for the implementation of national government policies to the extent that the policies relate to the county and in particular crop and animal husbandry; livestock sale yards markets); abattoirs; plant and animal diseases; and fisheries. It is the duty of the national government and county governments to provide an enabling environment for the development of the crop subsector. The national and county governments are to determine and promote the implementation of agricultural policies and measures in a manner designed to promote, support and enhance productivity in the sector³⁸.

Each county government is responsible for agricultural matters in accordance with Part 2 of Fourth Schedule. The National government, in accordance with Part 1 of Section 29 of the Fourth Schedule, is responsible for agricultural policy and for assisting the county governments on agricultural matters. Each county government should ensure uniformity and national standards in the agriculture sector, through legislation and administrative action, implement and act in accordance with national policy guidelines.

The national government has authority over policy issues, technical assistance, finance and training whereas the county governments have the responsibilities to do with financial management, priority setting, agricultural production and related extension services. Reports from various counties have indicated that there is increase in investment within the agricultural sector by 20-40% but there is lack of significant increase of agricultural productivity³⁹. The decline in agricultural productivity is due to poorly supported and unclear policy guidelines within the agricultural sector to achieve better growth, productivity, and food security. There is need for better usage of new standards and guidelines, appropriate coordination across both national and regional governments to misuse synergies and existing infrastructure for common development.

Kenya developed a County Programme Paper (CPP) on Ending Drought Emergencies (EDE) as part of the IGAD Drought Disaster Resilience and Sustainability Strategy. The paper presents

³⁸ See the Constitution of Kenya 2010

³⁹ <https://agricultureauthority.go.ke/index.php/en/resource-center/news/19-there-s-need-for-better-use-of-laws-to-boost-food-production>.

Kenya's system approach towards finishing intermittent drought crises. Under the initiative of the National Drought Management Authority (NDMA), the paper has been converted into a medium-term plan for implementing key segments of the ASAL strategy and Vision 2030's strategy for the development of northern Kenya and other arid terrains. The paper centers on establishing a more favorable environment for building drought resilience through building more grounded establishments and foundations for development. NDMA Strategic Plan 2018-2022 is the second NDMA Strategic Plan⁴⁰. The Plan expands on the experience of the primary Strategic Plan, and considers recent developments that have a course on drought risk management and climate change adaptation. It is informed by the needs of the Kenyan government and expresses strategies that add to the national development agenda, and the accomplishment of the SDGs and Agenda 2063 of the African Union, among other worldwide and regional commitments⁴¹.

The 2020-2025 Country Strategic Opportunities Programme (COSOP) is aligned with the (ASTGS) 2019-2020 six pillars. COSOP aims to strengthen the expected value chains and guarantee that poor country individuals take an interest in and advantage of them, utilizing climate resilient, comprehensive and favorable to pro-poor methodologies that improve Kenya's beneficial potential for people in the future. With minor adjustments, COSOP has retained its three strategic objectives which include: SO₁ ensures climate resilient and community-based natural resource management is improved; SO₂ ensures access to productivity-enhancing assets, rural finance, services and technologies is enhanced; SO₃ ensures sustainable access to improved post-production markets and technologies is improved.⁴²

⁴⁰ <https://www.ndma.go.ke/index.php/resource-center/policy-documents/send/44-policy-documents/5617-strategic-plan-2018-2022>

⁴¹ <https://www.ndma.go.ke/index.php/resource-center/policy-documents/category/44-policy-documents>

⁴² <https://www.ifad.org/en/document-detail/asset/40235700>

6. AGRICULTURE TRACKING, MONITORING AND REPORTING

Reinforcing the agri-food frameworks is key in the accomplishment of solutions for the current difficulties of poor nutrition and food insecurity in non-industrial nations. Improving food security and appropriate nutrition situation requires a multi-faceted methodology including taking a gander at alternative techniques and strategies to sustainable farming food production. To accomplish a world liberated from hunger and lack of healthy sustenance, it is crucial to provide sustainable solutions that give the biggest effects on development outcomes. Legitimate monitoring and evaluation tools and strategies assume a critical part in guaranteeing food security programs are planned, implemented/ actualized as a plan and that they improve the prosperity of the objective communities.

The primary goal of an early warning system is to monitor food security and nutritional status to notify decision-makers of forthcoming food deficiencies at national and local levels. The response that an early warning system triggers is a significant pointer of its success. The suitability of this response, thus, relies upon the connections between the data created by the system and the policymaking cycle. This linkage stays frail in a few sub-Saharan African nations. The ongoing food deficiencies experienced in Southern Africa and nations like Sudan and Ethiopia, are mostly because of the failure by governments and global associations to take public action of guaranteeing food security regardless of the presence of early warning frameworks in the nations. This is partly in light of the fact that food security monitoring frameworks, which are generally settled by external or benefactor funding, have not been completely coordinated into the decision making processes of the government. Not many endeavors have been made to characterize the criteria required for understanding the effect of the monitoring frameworks. All the more critically, there has not been any systematic assessment of the role of monitoring frameworks in the decision making process. Constant assessment of the advantages of food security monitoring systems is important to guarantee sustainability and permit them to mitigate food insecurity.

In order to advance urban farming, Nakuru County has begun the way toward establishing a board to monitor food security in the region. The proposed Nakuru County Urban Food Security and Agriculture Advisory Board is one of the features in the Urban Food and Agriculture 2019 Bill⁴³.

⁴³ <https://reliefweb.int/report/kenya/nakuru-county-forms-board-monitor-food-security>

The board will inform the executive on ways of promoting food security and agricultural exercises in the county. The Bill will uphold the approaches on urban agribusiness which incorporate developing, processing, and distribution of food.

7. AGRICULTURE FINANCE WINDOWS IN KENYA

Kenya has one of the highest agricultural productivity levels in the East African Region. 70% of the country's exports have an agricultural focus and the sector employs 85% of the rural workforce⁴⁴.

The 2020/2021 budget set aside Kshs.50.1 billion to agriculture and food sector. 8 billion of this went towards food security with focus on such projects as Kenya Climate Smart Agricultural Project, National Agricultural and Rural Inclusivity Project, Kenya Cereal Enhancement Programme, irrigation and land reclamation. However, the allocated budget contravenes the Malabo declaration in which countries committed to allocate at least 10% of the total budget to agriculture in order to achieve 6% growth of the agriculture economy. On the contrary, the budget for agriculture research was reduced by Kshs 950 million.⁴⁵ The budgetary allocation was slightly above 3 percent increase compared to 1.8 per cent in 2019/2020 fiscal year.⁴⁶ An additional funds were distributed as shown in the table below:

Item	Allocation in Kshs.
1. Subsidy to reach 200,000 small scale farmers through the e-voucher system	3 billion
2. Expand community household irrigation to cushion farmers from the adverse effects of weather and secure food supply chains	3.4 billion
3. Assist flower and horticultural farmers access international markets	1.5 billion
4. Enhance aquaculture business development projects	1.8 billion
5. Support small-scale irrigation and value addition	1.4 billion
6. Enhance resilience of pastoral communities	1.3 billion
7. Support processing and registration of title deeds	1.6 billion

⁴⁴ <http://www.invest.go.ke/agriculture/>

⁴⁵ <https://npck.org/the-2019-2020-kenya-budget-in-relation-to-agriculture/>

⁴⁶ <https://www.pd.co.ke/business/economy-and-policy/yatani-gives-sh61b-to-fire-up-struggling-agriculture-sector-40423/>

8. Advance agricultural loans through the Agriculture Finance Corporation	500 million
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The 2019/2020 budgetary allocation to agriculture sectors was as follows:

1. Crop diversification and revitalize the Miraa industry	1.0 billion
2. Rehabilitation of Fish landing sites	0.8 billion
3. Small-holder dairy commercialization	0.7 billion
4. Ongoing irrigation projects	7.9 billion
5. National Value Chain Support Programme	2.0 billion
6. Setting up the Coffee Cherry Revolving Fund	3.0 billion
7. Pay outstanding debts to sugar farmers for cane deliveries to public mills	0.7 billion

The Programme based budget for FY 2020/2021, the State Department for Crop Development and Agricultural Research which is a union of the former State Department for Crop Development and State Department for Agricultural Research tabled a total disbursement for the FY 2020/2021 Kshs. 40.1 billion⁴⁷. The 2020/21 budget policy statement proposed allocation is 4% less than that of 2019/20. 1.8% of the total BPS 2020/21 voted budget was allocated to food and nutrition⁴⁸. Moreover, the budgetary requirement of the ARUD area has been diminished by 50.6%. The 2020/21 BPS has not changed adverse patterns of low and declining allocations to the food and agricultural sector.

⁴⁷ <https://www.theelephant.info/data-stories/2020/08/07/kenyan-budget-allocation-in-the-sector-of-agriculture-for-the-fy-2020-2021/#:~:text=The%20budget%20allocation%20for%20the,billion%20allocated%20in%202020%2F2021.>

⁴⁸ <https://www.routetofood.org/wp-content/uploads/2020/06/RTFI-BPS-2020-Analysis.pdf>

8. CONCLUSIONS AND RECOMMENDATIONS: FUTURE OF AGRICULTURE AND FOOD SECURITY IN KENYA

From the study, it is evident that the Kenyan agricultural sector has a number of policies that are useful in the achievement of the SDGs 2 and 13 by 2030. The policies if well implemented can push the agricultural sector to greater heights while ensuring that climate change action is also considered. The only challenge comes on the downscaling of the policies and interpretation of some of the policies. There is a gap in policies that handle technology and its use within the sector.

This paper therefore recommends that:

- It is suggested that governments build up a new and extended policy agenda for agricultural augmentation and correspondence for rural development concentrating on food security and income generation of the rural poor.
- It is recommended that governments construct a platform to advance discourse and collaboration among applicable institutions and programs in all sectors with the point of building up an expansion and data service network for food security and income generation.
- Governments should activate institutional change within the public sector, aimed at supporting and promoting the new and expanded policy on extension and food security and the determinations instituted by the nationwide platform.
- It is recommended that the government to be fully involved in the improvement of the rain-fed agricultural enterprises both in ASALs and humid areas
- Measures addressing food security should involve all stakeholders and multi-dimensional both within the nation and outside the country.
- Agriculturalists should be enabled with knowledge on water system and off-season intensive farming of high-esteem crops.
- Formulation and implementation of agricultural extension policy is recommended to improve information flows and technology to the farming community. The extension officers should be enabled through budgetary allocation to visit the farmers and organize field days in their areas. Farmers exchange programs should be encouraged.

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